Sampling distributions and the bootstrap



Data science · University · Lecture 1

In this lecture

- Define a sampling distribution and SE.
- Use the bootstrap and know its limits.
- Sequence teaching simulation-first.

Remember

- Estimates vary from sample to sample.
- SE = σ/√n.
- Resampling can’t fix a biased sample.

Resample before you derive

1. Warm-up (5 min): What does 95% mean?
2. Simulate (20 min): 1,000 samples.
3. Derive (15 min): Explain the picture.
4. Bootstrap (15 min): One sample only.
5. Wrap-up (5 min): One-minute paper.

Let’s discuss

1. Which plot was narrower, and by how much?
2. Why can’t resampling fix a biased sample?
3. How will you assess interpretation, not just calculation?

Exit ticket

- Write one correct sentence interpreting a 95% confidence interval.